

Brunei ICT Usage and Expenditure Survey Report

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! Important

This is a sample report based on artificially generated data.

Introduction

The Brunei Darussalam ICT Usage and Expenditure Survey 2024 was conducted to understand the current landscape of information and communications technology (ICT) usage, focusing particularly on broadband internet usage and associated expenditure patterns among households across the country. This survey provides insights essential for policymaking, infrastructure improvement, and targeted ICT development initiatives.

Specifically, the objectives of the survey were:

- To identify broadband and data usage patterns across different demographic groups.
- To analyze monthly ICT expenditure and its relationship to usage.
- To evaluate spatial distribution and variability in ICT access and affordability.

Demography

This section provides an overview of the demographic characteristics of respondents who participated in the survey. The survey sampled a total of 2,000 respondents across four districts in Brunei Darussalam, comprising Belait (N=303), Brunei-Muara (N=1,453), Temburong (N=25), and Tutong (N=219).

Table 1: Demographic Characteristics of Survey Respondents

Demographic Characteristics	Overall N = 2,000 ¹	Belait N = 303 ¹	Brunei-Muara N = 1,45
Gender			
Male	990 (50%)	161 (53%)	702 (48%)
Female	1,010 (51%)	142 (47%)	751 (52%)
Age	39 (29, 47)	39 (29, 47)	39 (30, 48)
Education			
Primary School	38 (1.9%)	11 (3.6%)	22 (1.5%)
Lower Secondary	145 (7.3%)	20 (6.6%)	98 (6.7%)
O Level	422 (21%)	68 (22%)	298 (21%)
A Level	141 (7.1%)	23 (7.6%)	103 (7.1%)
National Certificate	59 (3.0%)	6 (2.0%)	47 (3.2%)
Diploma	52 (2.6%)	8 (2.6%)	37 (2.5%)
National Diploma	193 (9.7%)	23 (7.6%)	152 (10%)
Higher National Diploma	247 (12%)	36 (12%)	176 (12%)
Bachelor Degree	501 (25%)	74 (24%)	375 (26%)
Master Degree	185 (9.3%)	33 (11%)	131 (9.0%)
PhD	17 (0.9%)	1 (0.3%)	14 (1.0%)

¹n (%); Median (Q1, Q3)

²Pearson's Chi-squared test; Kruskal-Wallis rank sum test; Fisher's Exact Test for Count Data with simulated p-value (based on 2000 replicates)

```
dat |>
  select(district, gender, age, education) |>
  tbl_summary(by = district, label = list(
    "gender" ~ "Gender",
    "age" ~ "Age",
    "education" ~ "Education"
  )) |>
  add_overall() |>
  add_p(test.args = education ~ list(simulate.p.value = TRUE)) |>
  modify_header(label = "**Demographic Characteristics**")
```

Overall, there was a balanced gender distribution with 990 (50%) male and 1,010 (51%) female respondents. The proportions were consistent across districts, though notably, Temburong had a higher proportion of male respondents (64%) compared to other districts, but this difference was not statistically significant ($p = 0.2$).

The median age of respondents was 39 years, with an interquartile range (IQR) from 29 to 47 years. Age distribution showed minimal variability across districts, with median ages ranging between 38 and 42 years ($p = 0.7$).

Respondents represented a diverse range of educational backgrounds. The largest group had attained a Bachelor's degree (25%), followed by O Level qualifications (21%) and Higher National Diploma holders (12%). Other notable categories included National Diploma holders (9.7%), Master Degree holders (9.3%), and lower educational categories such as Lower Secondary (7.3%) and A Level (7.1%). Distribution across districts was relatively consistent, with minor differences not reaching statistical significance ($p = 0.5$).

This demographic snapshot indicates a representative and varied sample suitable for further analyses related to ICT usage and expenditure patterns.

Analysis of Survey Data

ICT Usage Patterns

Analysis reveals distinct patterns of broadband data usage among households in Brunei Darussalam. The distribution of broadband data usage (Figure 1) among survey respondents illustrates a clear right-skewed pattern, indicating most households consume moderate levels of data, typically ranging between approximately 150 GB and 500 GB monthly. The density curve reveals two prominent peaks, suggesting two common usage behaviors or subscription plans among respondents. The first peak around 200 GB likely corresponds to standard or moderate internet use, while the second peak near 400 GB may reflect heavier streaming, gaming, or multi-user household behaviors. There is a noticeable tail extending towards high data consumption levels (up to 1000 GB), though fewer households fall into this extreme usage category. This distribution highlights varied consumer behaviors and supports targeted broadband package planning and policy considerations.

```
dat |>
  ggplot(aes(q_fbusage)) +
  geom_histogram(
    aes(y = ..density..),
    binwidth = 100,
    fill = "steelblue",
    color = "black",
    alpha = 0.7
  ) +
  geom_density(aes(y = ..density..), color = "red", linewidth = 1) +
  labs(
    title = "Distribution of Broadband Data Usage",
    x = "Broadband Data Usage (GB)",
```

```
y = "Density"
)
```

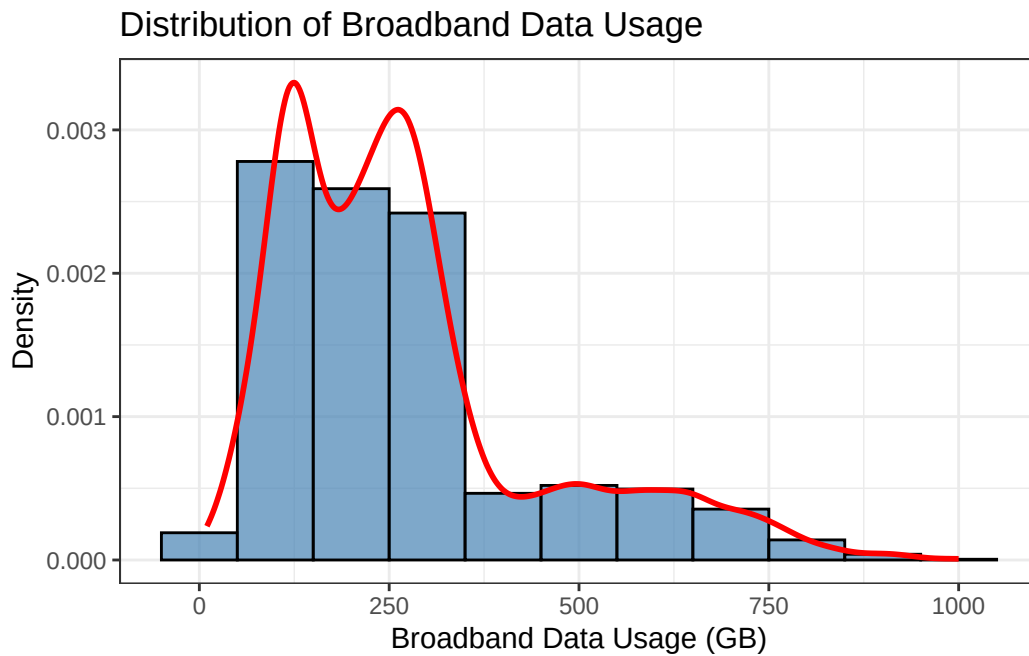


Figure 1: Distribution of fixed broadband data usage in Brunei Darussalam.

Figure 2: Broadband Data Usage by District

```
library(ggribes)
dat |>
  ggplot(aes(x = q_fbusage, y = district, fill = district)) +
  geom_density_ridges(scale = 5, alpha = 0.8, show.legend = FALSE) +
  scale_fill_viridis_d(option = "mako") +
  labs(
    title = "Broadband Data Usage by District",
    x = "Broadband Data Usage (GB)",
    y = "District"
  )
)
```

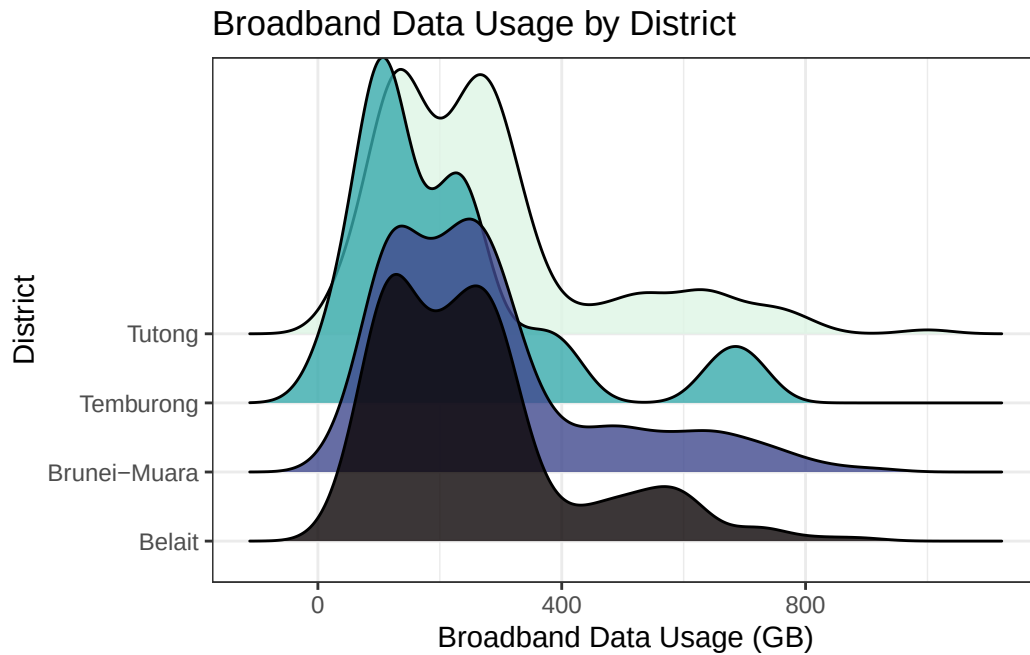


Figure 2: Broadband data usage by district.

Monthly ICT Expenditure

The survey also assessed monthly expenditure related to broadband and ICT services.

Table 2: Monthly ICT Expenditure Summary

Statistic	Expenditure (BND)
Minimum	[Placeholder]
Maximum	[Placeholder]
Mean	[Placeholder]
Median	[Placeholder]
Standard Deviation	[Placeholder]

Figure 3: Relationship between Monthly Expenditure and Data Usage

Placeholder for scatter plot with regression line: Monthly expenditure vs. Data usage

Expenditure Patterns by Education Level

Education levels were recategorized for analysis purposes. Differences in ICT spending habits across education levels provide insights into how demographic factors influence ICT affordability and access.

Figure 4: Monthly ICT Expenditure by Education Level

Placeholder for box plot or violin plot comparing ICT expenditure across education levels

Spatial Variability of ICT Expenditure

Geographic location has significant implications for broadband affordability and spending patterns.

Figure 5: ICT Expenditure by Mukim

Placeholder for spatial visualization or choropleth map showing ICT expenditure by Mukim

Conclusion

The Brunei ICT Usage and Expenditure Survey 2024 highlights critical differences in ICT usage and expenditure across demographic groups and geographic regions. Findings indicate targeted interventions could effectively address disparities and enhance broadband accessibility and affordability nationwide.

Future surveys should continue monitoring these trends, informing evidence-based policy decisions, and optimizing resource allocation for national ICT development.